

PROJECT TITLE

E-Learning Platform Analytics using MySQL

1. Project Overview & Objective

Project Overview

This project presents an end-to-end **SQL-based analysis of an e-learning platform**, focusing on learner behaviour, course performance, purchase activity, and revenue generation. A relational database was designed using MySQL to integrate learner, course, and transaction data and derive actionable business insights.

The project demonstrates the application of SQL for **data extraction, transformation, analytical analysis, learner segmentation, and performance evaluation**.

Objective

- To design and manage a structured relational database using MySQL.
 - To analyse learner purchasing behaviour and spending patterns.
 - To evaluate course and category-level performance.
 - To identify revenue-generating courses and high-value learners.
 - To detect underperforming and never-purchased courses.
 - To apply advanced SQL techniques for business-oriented analysis.
 - To generate insights that support **marketing, course promotion, and learner retention strategies**.
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2. Data Source

Source: Sample E-Learning Platform Dataset

Database: LearnersDB

Domain: Education / E-Learning Analytics

The dataset contains learner profiles, course information, and purchase transactions.

3. Problem Statement

The e-learning platform requires a structured approach to understand **which learners contribute the most revenue, which courses perform well, which categories have stronger demand, and where potential business opportunities exist**.

This project analyses transactional data to answer key business questions related to **revenue performance, learner value, course demand, and purchasing behaviour.**

4. Database Architecture

The database consists of three interconnected relational tables:

Learners

Stores learner identification and demographic information.

Courses

Contains course name, category, and pricing information.

Purchase

Records learner transactions, course purchases, quantities, and purchase dates.

Relationships:

Learners → Purchase ← Courses

Primary keys and foreign keys were implemented to maintain **referential integrity and relational consistency.**

5. Tools & Technologies

MySQL / SQL

- Database & Table Creation
- Data Insertion
- Data Retrieval
- Data Aggregation
- Relational Joins
- Subqueries
- Correlated Subqueries
- Common Table Expressions (CTEs)
- CASE Expressions
- NULL Handling
- SQL Views

6. Data Preparation & Validation

The database was structured and validated before analysis.

Activities Performed

- Created the relational database and tables.
 - Defined primary and foreign key relationships.
 - Inserted and validated sample transactional data.
 - Verified relationships between learners, courses, and purchases.
 - Identified courses without purchase records.
 - Applied NULL handling to ensure complete reporting.
 - Used aliases and formatted calculations for improved query readability.
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7. Analytical Approach

The analysis was performed across three levels:

- Total revenue
- Learner-wise spending
- Course purchase volume
- Category-wise revenue
- Unique learners
- Average revenue per purchase
- Top-performing courses

Investigated:

- High-value learners
- Learners purchasing across multiple categories
- Courses with low or zero purchase activity
- Categories generating higher revenue
- Learners spending above the overall average
- Spending compared with country-level averages

Based on observed purchasing patterns, the analysis identifies potential opportunities:

- High-value learners may have greater potential for repeat purchases.
- Popular courses and categories may continue to generate stronger demand.
- Learners purchasing multiple categories may be suitable for cross-selling.
- Low-performing courses may require targeted promotional strategies.

Note: Predictive insights are based on historical patterns in the sample dataset and do not represent machine-learning predictions.

8. Advanced SQL Implementation

The project demonstrates multiple SQL techniques:

JOIN Analysis

Implemented:

- INNER JOIN – combining related learner, purchase, and course information.
- LEFT JOIN – identifying all courses including those without purchases.
- RIGHT JOIN – demonstrating complete course-side retrieval.

Subqueries

Used to:

- Compare learner spending against average spending.
- Compare course prices against courses in a specific category.
- Compare learner spending with the average spending within their country.

CTE

A Common Table Expression was implemented to calculate **learner-level spending** and identify learners whose spending exceeded a defined threshold.

CASE Expression

Learners were segmented according to spending value:

- **Above 15,000 → High Value**
- **8,000–15,000 → Medium Value**
- **Below 8,000 → Low Value**

NULL Handling

IFNULL() was used to convert missing purchase counts into **0**, ensuring that courses without transactions were included in the analysis.

SQL View

A reusable view named **category_performance_view** was created to provide:

- Category
 - Total Revenue
 - Number of Purchases
 - Average Revenue per Purchase
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9. Key Business Insights

Learner Performance

The analysis identifies learners contributing higher revenue and provides a basis for **learner segmentation and targeted engagement**.

Course Performance

Purchase quantity analysis helps identify courses with stronger demand and provides insights for prioritising course promotion.

Category Performance

Category-level revenue analysis highlights the areas contributing more significantly to overall platform revenue.

Purchase Behaviour

Learners purchasing courses from multiple categories represent valuable opportunities for **cross-selling and personalised recommendations**.

Untapped Opportunities

Courses with no purchase activity can be investigated further for **pricing, visibility, content relevance, or promotional effectiveness**.

10. Business Recommendations

Based on the analysis, the following strategies are recommended:

1. Strengthen High-Performing Categories

Focus marketing and promotional campaigns on categories demonstrating stronger revenue and purchase demand.

2. Develop High-Value Learner Strategies

Offer personalised course recommendations, bundles, loyalty benefits, and premium learning opportunities to high-value learners.

3. Implement Cross-Selling

Recommend related courses to learners who have already purchased from multiple categories.

4. Improve Underperforming Courses

Review pricing, course positioning, descriptions, and promotional strategies for courses with low or zero purchase activity.

5. Establish Continuous SQL Reporting

Use reusable queries and the `category_performance_view` to monitor revenue and course performance regularly.

11. Conclusion

This project demonstrates an end-to-end application of **SQL for business and educational analytics**. By integrating learner, course, and transaction data, the analysis provides a structured view of **revenue performance, learner value, course demand, and purchasing behaviour**.

The implementation of **JOINS, aggregate functions, subqueries, correlated subqueries, CTEs, CASE expressions, NULL handling, and Views** demonstrates practical proficiency in SQL and relational database analysis.

The resulting insights can support **data-driven marketing, learner segmentation, course optimisation, cross-selling, and revenue growth strategies**.

Overall, the project showcases how SQL can transform transactional data into **meaningful business insights and actionable recommendations**.